

Lecture 1: Mathematical Foundations for Lattice-Based Cryptography

Groups, Rings, Fields, Modular Arithmetic,
Vector Spaces, Basis, and Lattices

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Lecture Notes — Session 1 (approx. 2 hours)

Purpose of this lecture. Every technical term used later in the thesis proposal — “lattice,” “modulus,” “ring $\mathbb{Z}_q[x]/(x^n + 1)$ ” — rests on a small set of algebraic building blocks. This lecture builds those blocks, one at a time, from the ground up: *Group* $\rightarrow$ *Ring* $\rightarrow$ *Field* $\rightarrow$ *Modular Arithmetic* $\rightarrow$ *Vector Space* $\rightarrow$ *Basis* $\rightarrow$ *Lattice*. Nothing here requires prior abstract algebra; every definition is followed immediately by concrete, worked examples.

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1 Why Start With Algebra?

Lattice-based cryptography is, at its core, about doing arithmetic on *vectors* whose coordinates live in a *finite, wraparound number system*, arranged according to strict algebraic rules. Before we can even state what a lattice is, we need precise answers to questions like:

- What does it mean for a set of numbers to have a well-behaved notion of “add” and “multiply”? (**Groups, Rings, Fields**)
- What happens when arithmetic “wraps around,” like a clock? (**Modulus**)
- What does it mean to build every point in a space out of a small number of building-block directions? (**Vector spaces, Basis**)
- What happens when we only allow *whole-number* combinations of those building blocks, instead of

any real-number combination? (**Lattice**)

This lecture answers each of these in order. By the end, the definition of a lattice (Section 7) should feel like the natural, inevitable next step, not an isolated new idea.

2 Sets and Binary Operations (Quick Refresher)

A **set** is just a collection of objects (numbers, vectors, symbols — anything). A **binary operation** on a set S is a rule that takes two elements of S and combines them into a third element, e.g., ordinary addition $+$ takes two numbers and produces their sum. We write $a * b$ for a general binary operation.

An operation is **closed** on S if combining two elements of S always gives another element of S (never something outside S). This sounds obvious but fails more often than you'd expect — see Example 1 below.

Example: Closure Can Fail

Let $S = \{1, 2, 3, 4, 5\}$ (just these five numbers) and let $*$ be ordinary addition. Then $2 * 4 = 6$, but $6 \notin S$. So ordinary addition is *not* closed on S . This is exactly the kind of problem modular arithmetic (Section 5) is built to fix.

3 Groups

Definition: Group

A set G together with a binary operation $*$ is called a **group**, written $(G, *)$, if all four of the following hold:

(G1) **Closure**: for all $a, b \in G$, $a * b \in G$.

(G2) **Associativity**: for all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.

(G3) **Identity**: there exists an element $e \in G$ such that $a * e = e * a = a$ for all $a \in G$.

(G4) **Inverses**: for every $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

If, in addition, $a * b = b * a$ for all $a, b \in G$ (the order doesn't matter), the group is called **abelian** (commutative).

Example: Groups You Already Know

- $(\mathbb{Z}, +)$ — the integers under addition. Identity is 0; the inverse of a is $-a$. This is an abelian group.
- $(\mathbb{R} \setminus \{0\}, \times)$ — nonzero real numbers under multiplication. Identity is 1; the inverse of a is $1/a$. Also abelian. (We had to remove 0 because 0 has no multiplicative inverse.)
- $(\mathbb{Z}, \times)$ — integers under multiplication — is **not** a group: 2 has no inverse in $\mathbb{Z}$ (we'd need $1/2$, which isn't an integer). This fails axiom (G4).

Remark: Why Cryptography Cares About Groups

Many classical cryptographic schemes (Diffie–Hellman, ElGamal, elliptic-curve cryptography) are built directly on top of group structure — the security relies on certain problems (like “given g and g^a , find a ”) being hard inside a specific group. Lattice-based cryptography instead builds on richer structures (rings and modules, coming next), but the group is always the first layer underneath.

Exercise: Try This Before Next Lecture

Is $(\mathbb{Z}_5 \setminus \{0\}, \times \bmod 5) = (\{1, 2, 3, 4\}, \times \bmod 5)$ a group? Check all four axioms directly by building the multiplication table. (You'll need Section 5 for what “mod5” means if it's new to you — come back to this after reading that section.)

4 Rings

A ring is what you get when you ask a set to support *two* operations — usually called “+” and “ $\times$ ” — that interact the way ordinary addition and multiplication do.

Definition: Ring

A set R with two binary operations $+$ and $\times$ is a **ring** $(R, +, \times)$ if:

(R1) $(R, +)$ is an abelian group (with identity element usually written 0).

(R2) $\times$ is associative: $(a \times b) \times c = a \times (b \times c)$.

(R3) The distributive laws hold: $a \times (b + c) = (a \times b) + (a \times c)$ and $(a + b) \times c = (a \times c) + (b \times c)$.

Note: a ring does *not* need multiplicative inverses, and multiplication need not be commutative in general (though in every ring we use in this thesis, it will be — these are called **commutative rings**). If there is an element $1 \in R$ with $a \times 1 = 1 \times a = a$ for all a , we say R is a **ring with unity**.

Example: Rings You Already Know

- $(\mathbb{Z}, +, \times)$ — the integers form a commutative ring with unity 1. Notice 2 still has no multiplicative inverse, and that's fine — rings don't require one.
- **Polynomial rings**, e.g. $\mathbb{Z}[x]$: the set of all polynomials with integer coefficients (like $3x^2 - x + 7$), with ordinary polynomial addition and multiplication. This is a ring for exactly the same reason $\mathbb{Z}$ is.

Looking Ahead: The Ring That Kyber and Dilithium Actually Use

Both Kyber and Dilithium do their arithmetic inside a specific ring of the form

$$R_q = \mathbb{Z}_q[x]/(x^n + 1),$$

read as: “polynomials with coefficients taken modulo q (Section 5), where we also reduce powers of x using the rule $x^n \equiv -1$.” This looks intimidating right now, but it is built from exactly the pieces in this lecture: a polynomial ring (this section), with coefficients from $\mathbb{Z}_q$ (Section 5), further “folded” by a fixed rule. We will unpack this fully in a later lecture — for now, just notice that “ring” is not an abstract curiosity; it's the literal object the cryptography is built on.

5 Fields

Definition: Field

A **field** is a commutative ring with unity F in which *every nonzero element has a multiplicative inverse* — that is, $(F \setminus \{0\}, \times)$ is also an abelian group. Informally: a field is a set where you can add, subtract, multiply, and divide (by anything except 0) and always stay inside the set.

Example: Fields and Non-Fields

- $\mathbb{Q}$ (rationals), $\mathbb{R}$ (reals), $\mathbb{C}$ (complex numbers) are all fields — you can divide by any nonzero element and stay inside the set.
- $\mathbb{Z}$ is **not** a field: $2 \in \mathbb{Z}$ has no multiplicative inverse in $\mathbb{Z}$ (again, $1/2 \notin \mathbb{Z}$). $\mathbb{Z}$ is a ring, but not a field.
- $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ under addition and multiplication mod 5 *is* a field — every nonzero element has an inverse (you checked this, or will check this, in Exercise 4.1). Fields built this way, from $\mathbb{Z}_p$ with p prime, are called **finite fields** or **Galois fields**, written $\mathbb{F}_p$ or $GF(p)$.

Remark: The One-Sentence Summary So Far

Group $\subset$ Ring $\subset$ Field, in the sense that a field is a ring with extra structure (division), and a ring is built from a group with extra structure (a second operation). Each step adds a capability: groups let you undo one operation (inverses); rings add a second, distributive operation; fields guarantee *both* operations are fully invertible (except division by 0).

6 The Modulus: Arithmetic That Wraps Around

Definition: Congruence Modulo n

For a fixed positive integer n (the **modulus**), two integers a and b are **congruent modulo n** , written

$$a \equiv b \pmod{n},$$

if n divides $(a - b)$ evenly — equivalently, a and b leave the same remainder when divided by n . The set of possible remainders, $\{0, 1, 2, \dots, n - 1\}$, together with addition and multiplication performed “with wraparound,” is written $\mathbb{Z}_n$ (or $\mathbb{Z}/n\mathbb{Z}$).

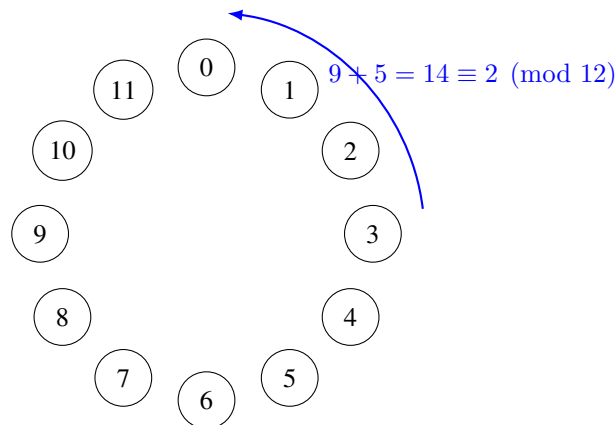


Figure 1: The clock analogy for modular arithmetic: $\mathbb{Z}_{12}$ works exactly like a 12-hour clock face. 9 o'clock plus 5 hours wraps around to 2 o'clock, i.e., $9 + 5 = 14 \equiv 2 \pmod{12}$.

Example: Computing Modulo

$17 \bmod 5$: divide 17 by 5 to get 3 remainder 2, so $17 \equiv 2 \pmod{5}$. Similarly, $23 \equiv 3 \pmod{10}$, and $-1 \equiv n - 1 \pmod{n}$ (e.g., $-1 \equiv 4 \pmod{5}$) — negative numbers wrap around too.

Remark: When Is $\mathbb{Z}_n$ a Ring? When Is It a Field?

$\mathbb{Z}_n$ is *always* a commutative ring with unity, for any $n \geq 2$. But it is a *field* only when $n = p$ is **prime**. If n is composite (e.g., $n = 6$), some nonzero elements fail to have multiplicative inverses — e.g., in $\mathbb{Z}_6$, there is no x with $2x \equiv 1 \pmod{6}$, since 2 shares a common factor with 6. This is exactly why cryptographic schemes that need a field (rather than just a ring) are built with a prime modulus q .

Looking Ahead: Why Cryptography Loves the Modulus

Modular arithmetic gives us two things cryptography needs badly: (1) a **finite** number system (so keys and computations have a fixed, bounded size, no matter how much you compute), and (2) built-in **one-wayness** in places — e.g., given only $a \bmod n$, you cannot tell which of infinitely many integers a actually was. The Learning-With-Errors (LWE) problem from the proposal document hides its secret inside equations that live in exactly this kind of finite, modular number system (there, the modulus is usually called q).

Exercise: Practice

(a) Compute $23 \times 15 \bmod 7$. (b) List all elements of $\mathbb{Z}_8$ that *do not* have a multiplicative inverse, and explain why (hint: compare each element to 8). (c) Confirm $\mathbb{Z}_7$ is a field by checking that 3 has a multiplicative inverse mod 7 (find x such that $3x \equiv 1 \pmod{7}$).

7 Vector Spaces, Basis, and Dimension

Definition: Vector Space

A **vector space** over a field F is a set V (whose elements are called **vectors**) equipped with vector addition and scalar multiplication (multiplying a vector by an element of F), satisfying the “obvious” rules you’d expect: addition makes $(V, +)$ an abelian group, and scalar multiplication distributes sensibly over both vector addition and field addition. The most important example for us: $\mathbb{R}^n$, the set of all n -tuples of real numbers, is a vector space over $\mathbb{R}$.

Definition: Linear Combination, Span, Linear Independence

Given vectors $v_1, \dots, v_k \in V$ and scalars $c_1, \dots, c_k \in F$, the vector $c_1v_1 + c_2v_2 + \dots + c_kv_k$ is called a **linear combination** of $v_1, \dots, v_k$. The set of *all* linear combinations of a set of vectors is called their **span**. The vectors $v_1, \dots, v_k$ are **linearly independent** if the only way to write $0 = c_1v_1 + \dots + c_kv_k$ is with every $c_i = 0$ — informally, no vector in the set is “redundant” (expressible using the others).

Definition: Basis and Dimension

A set of vectors $B = \{b_1, \dots, b_k\}$ is a **basis** of a vector space V if (1) B is linearly independent, and (2) B **spans** V (every vector in V can be written as a linear combination of B). Every basis of a given vector space has the same number of elements k , called the **dimension** of V .

Example: Basis of $\mathbb{R}^2$

The vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ form the “standard” basis of $\mathbb{R}^2$: any vector (a, b) is exactly $a \cdot e_1 + b \cdot e_2$. But this is far from the *only* basis — $\{(1, 1), (1, -1)\}$ is also a valid basis of $\mathbb{R}^2$ (check: they’re linearly independent, and any (a, b) can be written as a combination of them). A vector space almost always has infinitely many possible bases; some are more convenient to work with than others. **Keep this fact in mind — it becomes very important in Section 7.**

8 From Vector Spaces to Lattices

Everything so far has allowed *any* scalar from the field F (e.g., any real number) when forming linear combinations. A lattice is what happens when we impose one dramatic restriction: **only integer combinations are allowed.**

Definition: Lattice

Given n linearly independent vectors $b_1, \dots, b_n \in \mathbb{R}^n$ (called a **basis** of the lattice), the **lattice** they generate is the set of *all integer* linear combinations:

$$\mathcal{L}(b_1, \dots, b_n) = \left\{ \sum_{i=1}^n a_i b_i \mid a_i \in \mathbb{Z} \right\}.$$

Unlike a vector space (which is a continuous, solid region filling all of $\mathbb{R}^n$), a lattice is a **discrete** set of isolated points — it looks like an infinite, evenly-repeating grid, as in Figure 2. Formally, a lattice is a discrete additive subgroup of $\mathbb{R}^n$: it is itself a group under vector addition (check: it’s closed, has an identity $\vec{0}$, every point has an inverse, and addition is associative), but its elements do not fill up space continuously.

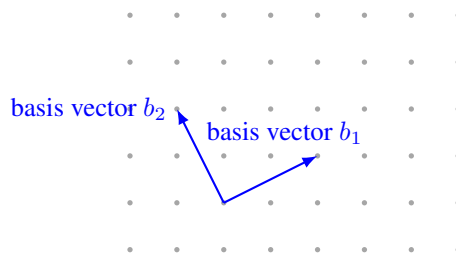


Figure 2: A 2-dimensional lattice generated by a basis $\{b_1, b_2\}$: every grid dot is reachable by some *integer* combination $a_1 b_1 + a_2 b_2$ (with $a_1, a_2 \in \mathbb{Z}$), starting from any chosen origin dot.

Remark: Same Lattice, Different Basis

Just as with vector spaces (Example 7), a single lattice has *many* different valid bases. Two bases generate the exact same set of grid points if and only if one can be converted to the other using an integer matrix with determinant ± 1 (a **unimodular** transformation) — we won’t need the details of this fact yet, but the consequence is critical: some bases consist of short, nearly-perpendicular vectors (a “good” basis, easy to compute with), while other, equally valid bases for the *same* lattice consist of long, nearly-parallel vectors (a “bad” basis, hard to compute with). **This gap between good and bad bases for the same lattice is the single most important idea in lattice-based cryptography:** the secret key is typically a good (short) basis, while the public key describes the same lattice using a bad (long, scrambled-looking) basis. Recovering the good basis from the bad one is believed to be extremely hard once the dimension n is large enough.

Example: A Tiny Lattice by Hand

Let $b_1 = (2, 0)$ and $b_2 = (0, 3)$ in $\mathbb{R}^2$. The lattice $\mathcal{L}(b_1, b_2)$ consists of every point $(2a_1, 3a_2)$ for integers a_1, a_2 — so it includes $(0, 0)$, $(2, 0)$, $(-2, 0)$, $(0, 3)$, $(2, 3)$, $(4, -3)$, and so on, but *not* $(1, 1)$ or $(1, 0)$, since those cannot be written as $(2a_1, 3a_2)$ for integer a_1, a_2 .

Looking Ahead: Where This Leads Next Lecture

With Groups $\rightarrow$ Rings $\rightarrow$ Fields $\rightarrow$ Modular arithmetic $\rightarrow$ Vector spaces $\rightarrow$ Basis $\rightarrow$ Lattice now in place, the next lecture can finally state the **Learning With Errors (LWE)** problem precisely: a secret vector, hidden inside noisy linear equations over $\mathbb{Z}_q$ (a ring, Section 5), which is exactly the kind of “bad basis” hardness described above, dressed up in a specific, standardized cryptographic form. We will also unpack the ring $R_q = \mathbb{Z}_q[x]/(x^n + 1)$ flagged in Section 4, since **module lattices** (lattices whose coordinates come from a ring like R_q instead of plain integers) are exactly what Kyber and Dilithium use in practice.

9 Summary Table

Structure	Requires	Example
Group	1 operation, inverses exist	$(\mathbb{Z}, +)$
Ring	2 operations, no division needed	$(\mathbb{Z}, +, \times), \mathbb{Z}[x]$
Field	Ring + division by nonzero elements	$\mathbb{Q}, \mathbb{R}, \mathbb{Z}_p$ (p prime)
Modulus	Wraparound arithmetic	$\mathbb{Z}_n = \{0, \dots, n-1\}$
Vector space	Field + vectors + scalar mult.	$\mathbb{R}^n$
Basis	Independent, spanning vector set	$\{(1, 0), (0, 1)\}$ in $\mathbb{R}^2$
Lattice	Vector space, but <i>integer</i> combos only	Grid of points in $\mathbb{R}^n$

10 Full List of Exercises (Assign as Homework Before Lecture 2)

Exercise: Homework Set

1. Verify $(\{1, 2, 3, 4\}, \times \bmod 5)$ is a group by writing out its full multiplication table (Exercise 4.1 above).
2. Compute $23 \times 15 \bmod 7$; list the non-invertible elements of $\mathbb{Z}_8$; verify $\mathbb{Z}_7$ is a field by finding the inverse of 3 (Exercise 6.1 parts a–c above).
3. Is $\mathbb{Z}_9$ a field? Why or why not? (Hint: think about what 9 is *not*.)
4. Show that $\{(1, 1), (1, -1)\}$ is a valid basis of $\mathbb{R}^2$ by (a) checking linear independence and (b) writing the vector $(4, 2)$ as a linear combination of them.
5. For the lattice generated by $b_1 = (1, 2)$ and $b_2 = (3, 1)$: is the point $(7, 5)$ in the lattice? (Find integers a_1, a_2 with $a_1b_1 + a_2b_2 = (7, 5)$, or show none exist.)
6. In your own words (3–4 sentences), explain why a “good basis” and a “bad basis” for the same lattice can make the same mathematical object feel completely different to work with computationally.